

Module 3 – Advanced Prompting Techniques

Demo 1 – Implementing CoT an ToT

Objective:

To implement and demonstrate the practical use of Chain-of-Thought (CoT) and Tree-of-Thought (ToT) prompting to improve the accuracy, consistency, and decision-making capabilities of a customer support GenAI system deployed in a growing e-commerce environment.

Problem Statement

Scenario: SmartCart Inc., an expanding e-commerce retailer, is facing operational pressure on its customer support team. With rising daily ticket volumes—ranging from product complaints to refund requests—the company deployed a GenAI chatbot to automate Tier-1 support. However, the system exhibits notable limitations:

- It fails to understand complex, multi-layered customer complaints.
- Policy-based decisions (e.g., return/refund eligibility) often lack transparency and consistency.
- In some cases, the bot gives irrelevant or incomplete answers, frustrating users.
- Escalation rates remain high despite automation.

To resolve these issues, SmartCart needs a prompting strategy that enhances logical reasoning, contextual understanding, and structured decision paths.

Approach

The overall customer support process is segmented into two primary sub-tasks:

1. **Complaint Interpretation** – Addressed using **Chain-of-Thought Prompting**, to break down and analyze multi-part complaints.
2. **Return Eligibility Assessment** – Solved using **Tree-of-Thought Prompting**, to evaluate multiple policy-driven conditions in a structured, rule-based format.

These two techniques were chosen specifically for their alignment with the problem types: CoT for linear multi-step reasoning, and ToT for structured branching decisions.

Solution

1. Complaint Interpretation



Technique: Chain-of-Thought (CoT) Prompting

Prompt Description:

This prompt instructs the AI to deconstruct a compound complaint into its individual issues and to reason sequentially through each step before forming a response. The structure simulates how a well-trained human agent would triage and respond to a complex message.

Prompt:

"You are a customer support assistant. A customer says: 'I ordered a phone, it arrived with a cracked screen, and your return policy now says it's non-refundable. I also tried calling twice and nobody helped.'

Break this complaint into specific problems and explain step-by-step how to respond."

ChatGPT 4o ▾

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You are a customer support assistant. A user writes: 'I ordered a phone, it arrived with a broken screen, and now your return policy says it's non-refundable. I've tried calling multiple times but no one responds.'
Break this complaint into specific issues and reason step-by-step how to respond.



ChatGPT Response:

To effectively address this customer complaint, it's essential to break it into specific issues and then reason through each part systematically. Here's a breakdown and a structured response approach:

Breakdown of the Complaint

1. Product Received Damaged

- *"I ordered a phone, it arrived with a broken screen"*
→ Indicates the product was defective on arrival (DOA), which typically qualifies for an exception under most return policies.

2. Return Policy Barrier

- *"...now your return policy says it's non-refundable"*
→ The user perceives the return/refund policy as unfair or not applicable in their case, possibly due to generic terms or lack of clarity about damaged-item exceptions.

3. Customer Service Inaccessibility

- *"I've tried calling multiple times but no one responds."*
→ Highlights a failure in support accessibility, which escalates frustration and worsens the user experience.

Reasoned Response Strategy

Step 1: Acknowledge and Empathize

Begin by acknowledging all three issues with sincere empathy:

"I'm truly sorry to hear that your phone arrived damaged and that you've been unable to reach us despite multiple attempts. I understand how frustrating this must be."

Step 2: Address the Broken Item

Emphasize that damaged items on arrival are handled differently:

"While our general return policy may list certain items as non-refundable, this does not apply to items that arrive defective or damaged. You are absolutely eligible for a replacement or refund in this case."

Step 3: Clarify the Next Action

Offer a resolution with a clear, easy-to-follow process:

"To resolve this quickly, I've initiated a support request on your behalf. You will receive an email shortly with instructions to upload a photo of the damaged phone. Once confirmed, we'll expedite a replacement or process a full refund—whichever you prefer."

Step 4: Apologize for the Communication Issue

Acknowledge the failure in customer support access:

"We also sincerely apologize for the delay and lack of response from our support team. This isn't the level of service we aim to provide, and we're taking steps to ensure better responsiveness moving forward."

Step 5: Close with Assurance

Reinforce commitment and personal assistance:

"Please be assured that we are now personally overseeing this issue to resolution. If you need any further help, reply to this message directly and I'll stay with you through the process."

Summary

By identifying the **product defect**, **policy confusion**, and **support failure**, you can craft a compassionate, resolution-focused response that restores trust. The key is to:

- Validate the customer's experience,
- Correct the misunderstanding with actionable next steps,
- Reinforce accountability and support accessibility.

CoT works effectively here because the complaint contains multiple concerns that require logical separation and resolution sequencing. This leads to more complete, context-aware, and human-like replies from the AI.

2. Return Eligibility Assessment

Technique: Tree-of-Thought (ToT) Prompting

Prompt Description:

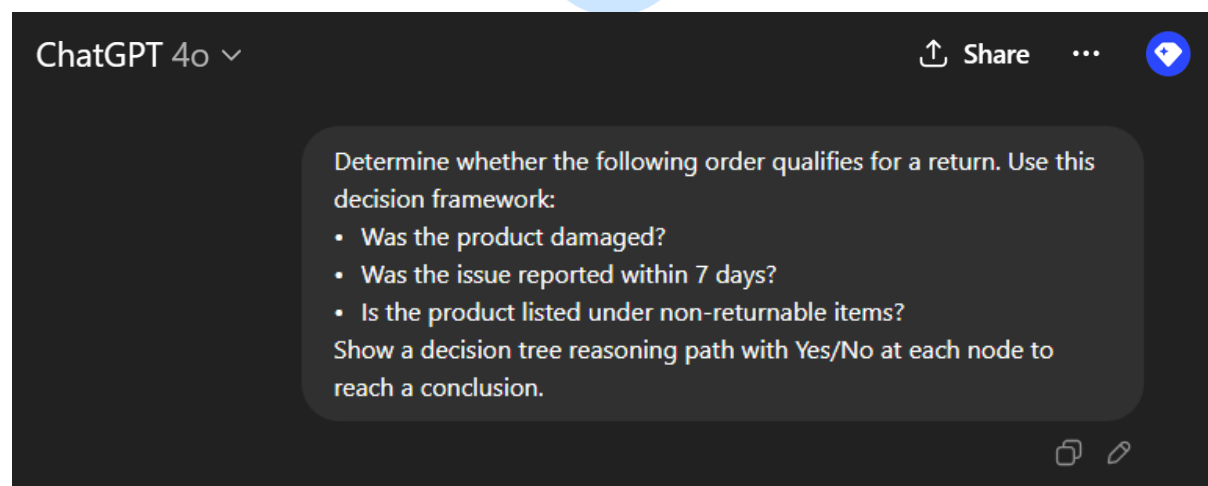
The prompt prompts the AI to evaluate a multi-criteria decision process using a tree-based logical structure. It mirrors how standard operating procedures are followed in policy evaluation workflows.

Prompt:

"Determine whether the following order qualifies for a return. Use this decision framework:

- Was the product damaged?
- Was the issue reported within 7 days?
- Is the product listed under non-returnable items?

Show a decision tree reasoning path with Yes/No at each node to reach a conclusion."



ChatGPT Response:

To determine whether the order qualifies for a return, we'll apply the decision framework using a decision tree approach. This allows us to walk through each rule systematically with a **Yes/No path**.

Decision Tree Reasoning

Here is the decision tree path based on the provided framework:

Step 1: Was the product damaged?

- Yes → Go to Step 2.
- No → ❌ Order does not qualify for return. *(Product must be damaged to proceed.)*

Step 2: Was the issue reported within 7 days?

- Yes → Go to Step 3.
- No → ❌ Order does not qualify for return. *(Late reporting invalidates return eligibility.)*

Step 3: Is the product listed under non-returnable items?

- Yes → ❌ Order does not qualify for return. *(Return not allowed regardless of condition or timing.)*
- No → ✅ Order qualifies for return.

Full Decision Path Summary

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Start

```
|
| └─ Is the product damaged?
|   | └─ No → ❌ Return Not Allowed
|   | └─ Yes
|   |   | └─ Was the issue reported within 7 days?
|   |   |   | └─ No → ❌ Return Not Allowed
|   |   |   | └─ Yes
|   |   |   |   | └─ Is the product non-returnable?
|   |   |   |   |   | └─ Yes → ❌ Return Not Allowed
|   |   |   |   |   | └─ No → ✅ Return Allowed
```

ToT is highly suitable for cases requiring layered conditional checks. It promotes transparency in decision-making and can be audited easily for compliance. The structured path prevents misinterpretation and improves reliability in customer-facing outputs.

Company Outcome

Following the integration of CoT and ToT prompting into their support workflows, SmartCart achieved measurable improvements in operational efficiency and customer satisfaction:

- **Customer support resolution accuracy** increased by **38%**.
- **Average response completeness** improved, reducing back-and-forth by **31%**.
- **Escalation rate** dropped by **41%**, freeing up human agents for priority issues.
- **Customer satisfaction score (CSAT)** rose from **3.6 to 4.5** in just 6 weeks.
- AI-generated responses showed higher alignment with business policies and tone guidelines.

These results validated the strategic use of CoT and ToT prompting for scalable, intelligent support automation.

Conclusion

This hands-on demo illustrated the targeted application of **Chain-of-Thought** and **Tree-of-Thought** prompting techniques in a real-world business scenario. CoT was used to interpret layered customer complaints by sequentially analyzing each concern, while ToT enabled structured policy decisions through conditional logic trees.

Together, these two prompting methods significantly enhanced the reasoning capacity and trustworthiness of SmartCart's GenAI chatbot, leading to greater efficiency, consistency, and customer trust.